

RUBEN MAXWELL

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Mathematics & Computer Science undergraduate at the University of Toronto with hands-on experience across software engineering, applied machine learning, and robotics. Proficient in Python, C++, and C#, with embedded robotics (VEX/Arduino), model development in PyTorch and TensorFlow (CUDA/GPU), and agentic AI coding tools. Builds production backends, real-time event-driven systems, computer-vision models, and LLM/RAG applications.

EDUCATION

University of Toronto — Honours B.Sc. Toronto, ON • 2025-2029

TECHNICAL SKILLS

Languages: Python, C++, C#, C, Java, JavaScript/TypeScript, SQL, Bash
Machine Learning & AI: PyTorch, TensorFlow, Keras, CUDA / GPU Acceleration, NumPy, Large Language Models (LLMs), Fairlearn, Qiskit, PennyLane, Cirq
AI Engineering & Agents: Claude Code, Aider, Codex CLI (agentic coding workflows), LangGraph, LlamaIndex, CrewAI, RAG pipelines, Qdrant, Chroma, Weaviate
Robotics & Embedded: VEX Robotics, Arduino, Raspberry Pi, Unity (C#), Embedded C++, Sensor Integration & Motor Control
Tools & Systems: Linux, Docker, Git/GitHub, CI/CD, REST APIs, RabbitMQ, AWS
Databases: PostgreSQL, MongoDB, MySQL, Firebase

EXPERIENCE

Software Engineer (Innovation Lab) — Fourth Dimension (4D) Toronto, ON • May 2026 - Present

- Built and deployed containerized microservices with Docker, standardizing environment setup and improving deployment reliability across distributed backend systems.
- Engineered real-time, event-driven workflows with RabbitMQ and optimized MongoDB queries and indexing, measurably reducing request latency and improving throughput.
- Developed and maintained backend REST APIs in an agile team, shipping 4+ production releases, using agentic coding tools (Claude Code, Aider) to accelerate implementation and refactoring.

Software Developer & Project Manager — Qmorphix LLC Toronto, ON • Jul 2025 - Sep 2025

- Applied machine learning in Python to MRI image reconstruction and denoising, comparing VQE, QNN, and quantum-kernel methods on noisy medical-imaging data.
- Built and benchmarked simulation pipelines with Qiskit, PennyLane, and Cirq, improving simulation accuracy and reducing model training time.
- Ran controlled experiments across 100+ datasets and coordinated project deliverables, substantially improving reconstruction quality and reducing noise impact.

Frontend Developer — My Code Club Mississauga, ON • Jul 2025 - Aug 2025

- Led a responsive redesign of the company website (HTML, CSS, JavaScript, Tailwind CSS) and restructured its architecture, improving cross-device usability and cutting content-update time; deployed via Firebase.

AI Ambassador — Inspirit AI Sep 2023 - Jul 2025

- Represented a Stanford/MIT-mentored AI program as a student ambassador, mentoring peers on introductory Python and machine-learning concepts.

PROJECTS

Robotics & Embedded Systems — VEX & Arduino C++, Embedded

- Designed, built, and programmed VEX robots and Arduino microcontrollers in C++, implementing closed-loop motor control, real-time sensor feedback, and both autonomous and driver-controlled routines on physical hardware.

Skin Cancer Detection — URSA Case Competition Python, TensorFlow/Keras

- Built a computer-vision pipeline, training GPU-accelerated CNNs on the HAM10000 dataset (10,015 images) for melanoma-vs-nevus classification, reaching ~0.90 AUC and 89-91% validation accuracy.
- Audited model fairness across patient groups with Fairlearn, surfacing a 4.45% accuracy gap and false-negative-rate disparity by sex.

Decyp3r — Cross-Platform Mobile App React Native, Expo, TypeScript, Firebase

- Built a daily microgame app (iOS/Android/Web) with 6 games, streak tracking, and live Firestore leaderboards; achieved 95%+ crash-free sessions and transactional, anti-spam score submission.

ForgeFountain — Real-Time Market Opportunity Analyzer APIs, Data Pipeline

- Built a live market/recipe analyzer with caching, fallbacks, and retry handling for reliability under API rate limits, cutting manual price-checking time and improving margins.

CERTIFICATIONS & HONORS

Certifications: Machine Learning with Python — IBM • AI For Everyone — DeepLearning.AI / Coursera • Version Control — Meta
Honors: 3× Hackathon Winner • Sir Isaac Newton Physics Exam — Top 500 worldwide of 3,808 (Waterloo, 2025) • Top 5 — NASA SpaceHacks • Languages: English (Native), French (Working)